DATABASE SYSTEMS IMPLEMENTATION

Hair Fix Salon
SQL Test Report

Executable acceptance-test subset for the online appointment booking database.

Prepared by	Robert Martinez-Olivera and Yaswanth Gosukonda

Execution date	October 25, 2026
Result	10 executed: 7 PASS, 3 FAIL
Target platform	MySQL 8.x

1. Scope and execution environment

The test plan in the Functional Requirements Document and Technical Design Document contains 20 acceptance cases. The five supplied SQL files implement tables, keys, checks, indexes, and views. They do not include seed data, stored procedures, triggers, database users, or role grants. This report executes the ten cases that can be meaningfully tested against the supplied schema and lists the remaining cases as not executed.

Local execution used an isolated MariaDB 12.3.3 server because MySQL 8 was not installed in the available environment. The MySQL-only collation utf8mb4_0900_ai_ci was changed in memory to utf8mb4_unicode_ci for this local compatibility run. The supplied SQL file itself was not changed. The group must repeat the build in MySQL 8 / MySQL Workbench before final submission.

2. Build verification

Object	Count	Result
Base tables	14	Created successfully
Views	7	Created successfully
Named performance indexes	8	Created and available
CHECK constraints	16	Present
Foreign keys	22	Present
Primary keys	14	Present
UNIQUE constraints	8	Present
Stored procedures	0	Not supplied
Triggers	0	Not supplied

3. Executed acceptance tests

Test	Status	Actual result
TC-01	PASS	Duplicate email was rejected.
TC-03	FAIL	Overlapping appointment was accepted by direct INSERT.
TC-07	PASS	Booked snapshot remained 50.00 and 60 minutes.
TC-08	FAIL	Ordinary AppointmentHistory UPDATE was accepted.
TC-14	PASS	Net revenue was 80.00: 100.00 charge minus 20.00 refund.
TC-15	FAIL	Inactive-service booking was accepted by direct INSERT.
TC-16	PASS	Appointments 1 and 2 returned in start-time order.
TC-17	PASS	EndTime before StartTime was rejected.
TC-18	PASS	Blank phone and invalid preference were rejected.
TC-20	PASS	Final-status payments saved; statuses stayed unchanged.

Summary: 10 tests executed, 7 passed, and 3 failed. A failed test means the current supplied schema does not enforce the documented expected behavior.

4. Supporting query evidence

Monthly revenue view

Month	Net	Succeeded charges	Succeeded refunds	All rows
2026-10	80.00	1	1	4

Daily schedule view for staff 1 on October 27, 2026

ID	Start	Customer	Service	Staff	Status
1	09:00	Taylor Reed	Haircut	Jordan Lee	Scheduled
2	11:00	Taylor Reed	Haircut	Jordan Lee	Completed

EXPLAIN selected idx_Appointment_StaffStartStatus for the staff/date query. The access type was range, the estimate was two rows, and the Extra value was Using where; Using index.

5. Failed-test interpretation

- TC-03 failed because no booking procedure, trigger, exclusion constraint, or other database object checks appointment time overlap.
- TC-08 failed because no trigger or database privilege prevents ordinary UPDATE or DELETE operations on AppointmentHistory.
- TC-15 failed because ActiveFlag is stored but no controlled booking procedure checks it before inserting an appointment.

These failures are consistent with the supplied implementation. They should be reported, not converted into passes, until the missing procedures, triggers, and permissions are implemented.

6. Test cases not executed

Test	Reason not executed
TC-02	Controlled booking and automatic history object not supplied.
TC-04	Status authorization procedure and database grants not supplied.
TC-05	Controlled status-transition procedure not supplied.
TC-06	Booking transaction and forced-failure mechanism not supplied.
TC-09	Controlled rescheduling procedure not supplied.
TC-10	Rescheduling procedure with 24-hour rule not supplied.
TC-11	Availability object that subtracts blocked time not supplied.
TC-12	Controlled payment-recording procedure not supplied.
TC-13	Payment authorization procedure and grants not supplied.
TC-19	Controlled rescheduling procedure not supplied.

7. Structural findings

- PaymentTransaction has only its primary key. The supplied keys.sql omits its foreign keys, and checks.sql omits amount, type, and status checks plus the unique external-reference constraint described by the design.
- No procedures or triggers are present. Therefore overlap prevention, workflow authorization, atomic booking/rescheduling, immutable history, and inactive-service validation are not implemented in the supplied files.
- No standalone initial-data script was supplied. The test script provides a small repeatable test seed, but the team should decide whether it is suitable as the official project seed.

8. Recommended next action

Run the combined schema build and test script in MySQL 8 / MySQL Workbench. Capture screenshots of the final TestResults table, the revenue output, the daily schedule output, and the EXPLAIN result. Keep the three failures and ten not-executed cases clearly labeled unless the missing implementation is added and those cases are rerun.